

Adrian Diaz

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EDUCATION

California State University, Fullerton

Bachelor of Science in Computer Science

Fullerton, CA

Expected Graduation Date: May 2026

EXPERIENCE

Data Entry

April 2024 – Present

Smile Brands Inc.

Costa Mesa, CA

- Configured dental insurance plans via Dentrrix Ascend, ensuring accurate system integration and compliance with office submitted tickets.
- Delivered prompt technical assistance to 600+ offices, diagnosing and resolving issues related to insurance configurations, fee schedule uploads, and data integrity.
- Implemented data validation protocols using advanced Excel functions to identify and resolve data inconsistencies, improving fee schedule accuracy by 25%.

Software Engineer Intern

July 2021 – August 2021

University of California, Irvine

Irvine, CA

- Developed Python code for autonomous motion planning, modeling navigation logic and pathfinding algorithms.
- Implemented Dijkstra's algorithm to improve route planning efficiency for obstacle rich environments.
- Visualized algorithm behavior using Matplotlib and mathematical libraries to identify performance bottlenecks and enhance motion logic.
- Contributed to detailed technical documentation on algorithm design, facilitating team knowledge sharing.

PROJECTS

Azure Cloud Migration Project | *Microsoft Azure, Windows Server, Active Directory*

March 2026 – Present

- Migrated a Windows Server 2022 Active Directory to Microsoft Azure, configuring virtual networking, access controls, and authentication services.
- Performed pre-migration security hardening including system patching and service / port reduction.
- Designed cloud architecture using Azure Virtual Networks, subnets, and network security groups to isolate infrastructure and enforce least privilege access control.
- Developed a migration plan including risk assessment, cost considerations, and a rollout strategy for scaling enterprise infrastructure across 20 virtual machines.

Accountabuddy | *React Native, Firebase, TypeScript*

January 2026 – Present

- Built a mobile application that improves task completion rates by introducing social accountability through shared task management.
- Engineered social features enabling users to add friends, share tasks, and track progress across 100+ active users.
- Utilized Firebase to manage backend functionality, enabling task updates, commenting, and user interactions.

Rubix Tutor | *Python, NumPy, Matplotlib*

August 2025 – January 2026

- Implemented an AI based Rubik's Cube solver using IDA* and IDDFS to search a large state space and generate optimal move sequences.
- Designed an admissible heuristic with a corner orientation pattern database, decreasing average solve depth.
- Modeled cube corner orientation and precomputed a pattern database with BFS to guide search.
- Developed a visualization interface with Matplotlib to display cube states and AI recommended moves.

Logfolio | *HTML, CSS, JavaScript, Node.js, Express.js, MongoDB*

June 2025 – August 2025

- Developed a full stack trade journaling web application to log, track, and analyze stock trades.
- Integrated external APIs (Finnhub, NewsAPI, YahooFinance) to provide market data and IPO events.
- Deployed the application using Render and MongoDB Atlas, ensuring scalability and reliability.

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript, Python, C++, Bash, SQL

Frameworks / Libraries: Node.js, Express.js, React, React Native, NumPy, Matplotlib

Databases: Firebase, MySQL, MongoDB

Developer Tools: Microsoft Azure, Git, GitHub, Postman, MongoDB Compass, Vercel, Visual Studio Code, Linux